SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 4

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

A book has n typos. Two proofreaders, Prue and Frida, independently read the book. Prue catches each typo with probability p_1 and misses it with probability $q_1 = 1 - p_1$, independently, and likewise for Frida, who has probabilities p_2 of catching and $q_2 = 1 - p_2$ of missing each typo. Let X_1 be the number of typos caught by Prue, X_2 be the number caught by Frida, and X be the number caught by at least one of the two proofreaders.

(a) Find the distribution of X.

(b) For this part only, assume that $p_1 = p_2$. Find the conditional distribution of X_1 given that $X_1 + X_2 = t$.

Solution

(a) The probability that both proofreaders miss a typo is

$$q_1 q_2 = (1 - p_1)(1 - p_2).$$

Thus,

$$X \sim \text{Bin}(n, 1 - q_1 q_2),$$

with

$$P(X = k) = \binom{n}{k} (1 - q_1 q_2)^k (q_1 q_2)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

(b) Let $p_1 = p_2 = p$ and write $q = 1 - p$. Since X_1 and X_2 are independent,

$$X_1 \sim \text{Bin}(n, p), \qquad X_2 \sim \text{Bin}(n, p), \qquad X_1 + X_2 \sim \text{Bin}(2n, p).$$

Given $X_1 + X_2 = t$, we have

$$\begin{aligned} P(X_1 = k \mid X_1 + X_2 = t) &= \frac{P(X_1 = k,\ X_2 = t - k)}{P(X_1 + X_2 = t)} \\ &= \frac{P(X_1 = k) P(X_2 = t - k)}{P(X_1 + X_2 = t)} \\ &= \frac{\binom{n}{k}(1-q)^k q^{n-k} \binom{n}{t-k}(1-q)^{t-k} q^{n-t+k}}{\binom{2n}{t}(1-q)^t q^{2n-t}} \\ &= \frac{\binom{n}{k}\binom{n}{t-k}}{\binom{2n}{t}}. \end{aligned}$$

Therefore,

$$P(X_1 = k \mid X_1 + X_2 = t) = \frac{\binom{n}{k}\binom{n}{t-k}}{\binom{2n}{t}}, \qquad 0 \le k \le \min(n, t).$$

Problem 2

Let X be a random day of the week, coded so that Monday is 1, Tuesday is 2, *etc.* (so X takes values $1, 2, \ldots, 7$ with equal probabilities). Let Y be the next day after X. Do X and Y have the same distribution? What is $P(X < Y)$?

Solution

We have

$$P(X = k) = \frac{1}{7}, \qquad P(Y = k) = \frac{1}{7}, \qquad k = 1, 2, \ldots, 7.$$

Thus, X and Y have the same distribution. The possible pairs are

X	1	2	3	4	5	6	7
Y	2	3	4	5	6	7	1

The inequality $X < Y$ holds for the first six pairs, so

$$P(X < Y) = \frac{6}{7}.$$

Problem 3

For X and Y binary digits (0 or 1), let $X \oplus Y$ be 0 if $X = Y$ and 1 if $X \neq Y$ (this operation is called exclusive or (often abbreviated to XOR), or addition mod 2).

(a) Let $X \sim \text{Bern}(p)$ and $Y \sim \text{Bern}(1/2)$, independently. What is the distribution of $X \oplus Y$?

(b) With notation as in sub-problem (a), is $X \oplus Y$ independent of X? Is $X \oplus Y$ independent of Y? Be sure to consider both the case $p = 1/2$ and the case $p \neq 1/2$.

(c) Let $X_1, ..., X_n$ be i.i.d. (i.e., independent and identically distributed) Bern(1/2) R.V.s. For each nonempty subset J of $\{1, 2, ..., n\}$, let

$$Y_J = \oplus_{j \in J} X_j.$$

Show that $Y_J \sim \text{Bern}(1/2)$ and that these $2^n - 1$ R.V.s are pairwise independent, but not independent.

Solution

(a) Let $Z = X \oplus Y$. The truth table is

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

Since $X \sim \text{Bern}(p)$ and $Y \sim \text{Bern}(1/2)$ independently,

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p, \qquad P(Y = 1) = P(Y = 0) = \frac{1}{2}.$$

Hence,

$$P(Z = 1) = P(X = 0, Y = 1) + P(X = 1, Y = 0) = \frac{1}{2}(1 - p) + \frac{1}{2}p = \frac{1}{2},$$
$$P(Z = 0) = P(X = 1, Y = 1) + P(X = 0, Y = 0) = \frac{1}{2}p + \frac{1}{2}(1 - p) = \frac{1}{2}.$$

Thus, $X \oplus Y \sim \text{Bern}(1/2)$.

(b) If X is known, then Z is either Y or $1 - Y$. In either case, the probabilities of $Z = 0$ and $Z = 1$ remain $1/2$. Therefore, $X \oplus Y$ is independent of X.

If Y is known, then

$$P(Z = 1 \mid Y = 0) = P(X = 1) = p, \qquad P(Z = 1 \mid Y = 1) = 1 - p.$$

(i) If $p = 1/2$, then $X \oplus Y$ is independent of Y.

(ii) If $p \neq 1/2$, then $X \oplus Y$ is not independent of Y.

(c) For a nonempty subset J, we have

$$Y_J = \bigoplus_{j \in J} X_j.$$

Fix all the variables in Y_J except one, X_j. Then Y_J is either X_j or $1 - X_j$. Since $X_j \sim \text{Bern}(1/2)$, it follows that $Y_J \sim \text{Bern}(1/2)$.

Let J and K be two distinct nonempty subsets. After interchanging their names if necessary, choose an index $j \in J \setminus K$. Fix all the variables except X_j. Then Y_K is fixed, while Y_J is either X_j or $1 - X_j$. Since $X_j \sim \text{Bern}(1/2)$, Y_J still has distribution $\text{Bern}(1/2)$. Thus, knowing one of Y_K and Y_J does not change the distribution of the other, so they are pairwise independent.

However, they are not mutually independent. Consider X_1, X_2, and $X_3 = X_1 \oplus X_2$; these variables are dependent.

Problem 4

Let X be a discrete R.V. whose distinct possible values are $a_1, a_2, ..., a_n$, with probabilities $p_1, p_2..., p_n$, respectively (so $p_1 + p_2 + \cdots + p_n = 1$). The entropy of X is defined to be the average surprise of learning the value of X:

$$H(X) = \sum_{j=1}^{n} p_j \log_2(1/p_j).$$

Show that the maximum entropy for X is when its distribution is uniform over $a_1, a_2, ..., a_n$, i.e., $p_j = 1/n, \forall j$.

Solution

Assume $p_j > 0$. For the concave function $f(x) = \log_2 x$, Jensen's inequality gives

$$\sum_{i=1}^{n} a_i f(x_i) \leq f\left(\sum_{i=1}^{n} a_i x_i\right).$$

Therefore,

$$H(X) = \sum_{j=1}^{n} p_j \log_2\left(\frac{1}{p_j}\right) \leq \log_2\left(\sum_{j=1}^{n} p_j \frac{1}{p_j}\right) = \log_2 n.$$

Equality holds if and only if

$$p_j = \frac{1}{n}, \qquad \forall j.$$